



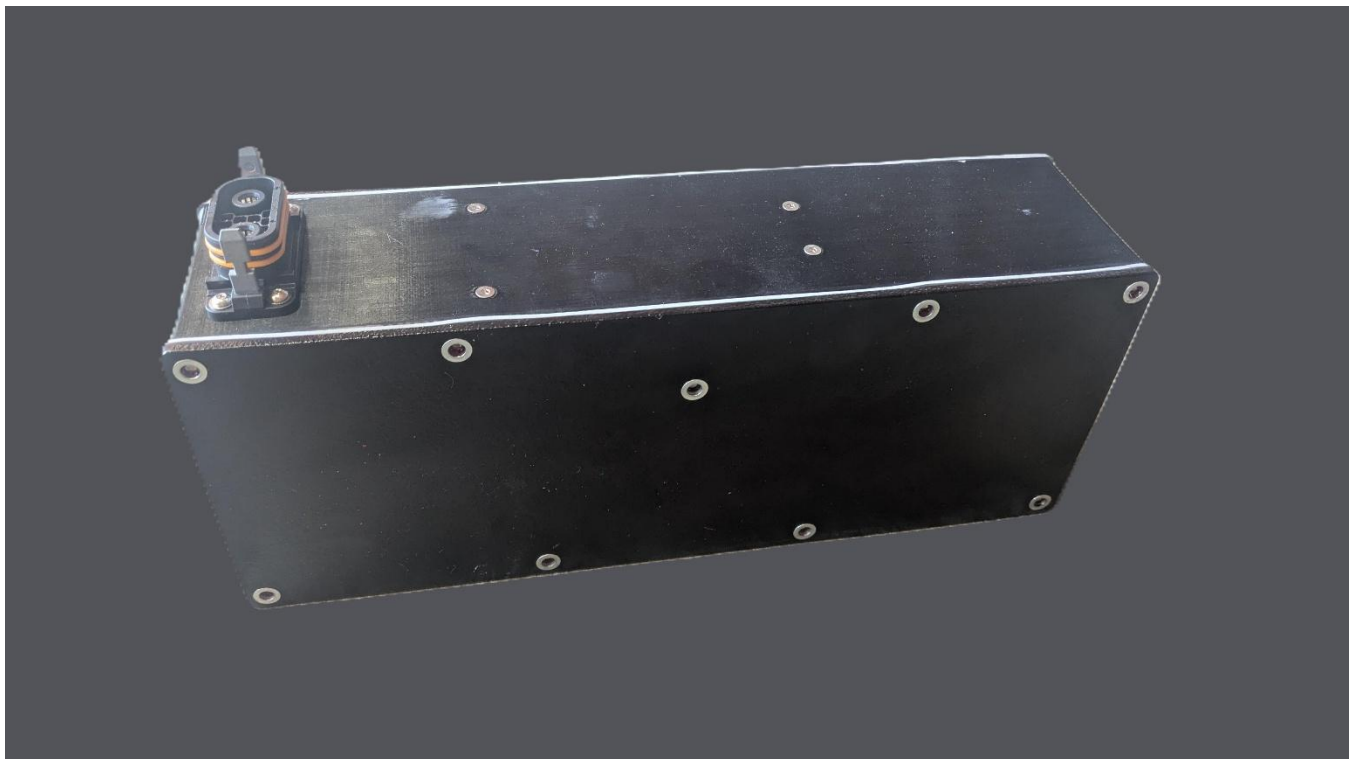
Falcon 15 EB Battery

Product Manual | Rev 1.0 | 05/19/2026

The Morelle Battery is a high-silicon NCM 21700 lithium battery pack in a 13S3P configuration with an integrated battery management system (BMS). The BMS monitors cell voltages, temperatures, and protection states in real-time, and communicates over CAN bus for integration with chargers or host controllers.

KEY FEATURES

- Real-time cell voltage and temperature monitoring
 - Passive cell balancing
 - Charge and Discharge MOS protection
 - CAN bus communication for battery (250 kbps, 29-bit extended frame IDs)
 - Morelle custom fast charge protocol (up to 50A)
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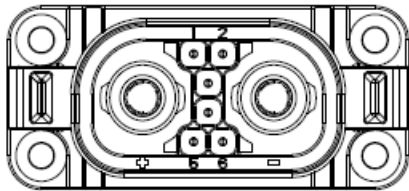
BATTERY PACK SPECIFICATIONS

	Parameter	Requirements
Cell and Pack Basic Spec	Nominal Voltage	46.8V
	Charge Voltage	54.5V
	Cell Type	High Silicon - NCM 21700
	Charge Mode	Morelle's custom Fast charge protocol
	Pack Method	13S3P
	Capacity at 25C, 0.2C discharging rate	15Ah (Nominal), 14.5Ah (Min)
	Standard Charge Current	1C (15A)
	Maximum Charge Current	5C (70A)
	Standard Discharge Current	1C (15A)
	Maximum Discharge Current / Duration	1.0C (15A) / 4 hours
		3.0C (45A) / 1 minute
		6.5C (100A) / 1 second
	Ambient (storage) Temperature limits	-10 – 45C
	Charging temperature	0C ~ 55C (0C-15C limit current to 18A)
	Discharge Temperature	-10C ~ 60C
	Cycle Life	>800 cycles
	Voltage Cutoff	35V
Size and Weight	Target Max Size (L X W X H)	310 x 81 x 113.5 mm
	Target Weight	5 kg
BMS	BMS required	Yes
	Cell Balancing	Passive
	Indicators	No
	Communication	CAN
	On Switch	Yes – Through charge port to exterior switch
	BMS current consumption in shutdown	< 45 uW
Protection Coverage	Pack Level Overvoltage Protection	Level 1: 54.7V Delay: N/A alarm only Recovery: N/A Level 2: 55.3V Delay: 500 mS Recovery: 54.5V Level 3: 55.9V Delay: 500 ms Recovery: 55.0V
	Pack Level Undervoltage Protection	Level 1: 34.8V Delay: N/A alarm only Recovery: N/A Level 2: 33.8V Delay: 500 mS Recovery: 35.0V Level 3: 32.5V Delay: 500 ms Recovery: 34.0V
	Cell Level Overvoltage Protection	Level 1: 4.22V Delay: N/A alarm only Recovery: N/A Level 2: 4.25V Delay: 500 mS Recovery: 4.20V Level 3: 4.30V Delay: 500 ms Recovery: 4.25V .
	Cell Level Undervoltage Protection	Level 1: 2.64V Delay: N/A alarm only Recovery: N/A Level 2: 2.54V Delay: 500 mS Recovery: 2.70V Level 3: 2.50V Delay: 500 ms Recovery: 2.60V ..
	Pack Level Overcurrent Protection for Charging	Level 1: 72A Delay: N/A alarm only Recovery: N/A Level 2: 75A Delay: 500 mS Recovery: Disconnect Chg Level 3: 78A Delay: 500 ms Recovery: Disconnect Chg
	Pack Level Overcurrent Protection for Discharging	Level 1: 72A Delay: N/A alarm only Recovery: N/A Level 2: 90A Delay: 500 mS Recovery: Remove Load Level 3: 100A Delay: 500 ms Recovery: Remove Load
	Short Circuit Protection	>200A Dely: 500 mS Recovery: Fuse replacement

Overtemperature Protection for Charging	Level 1: 55C Delay: N/A alarm only Recovery: N/A Level 2: 60C Delay: 500 mS Recovery: 45 C Level 3: 65C Delay: 500 ms Recovery: 45 C
Overtemperature Protection for Discharging	Level 1: 60C Delay: N/A alarm only Recovery: N/A Level 2: 63C Delay: 500 mS Recovery: 55 C Level 3: 65C Delay: 500 ms Recovery: 55 C
Undertemperature Protection for Charging	Level 1: 0C Delay: N/A alarm only Recovery: N/A Level 2: -3C Delay: 500 mS Recovery: 0 C Level 3: -5C Delay: 500 ms Recovery: 0 C
Undertemperature Protection for Discharging	Level 1: -10C Delay: N/A alarm only Recovery: N/A Level 2: -13C Delay: 500 mS Recovery: -10 C Level 3: -15C Delay: 500 ms Recovery: -10 C
BMS Overtemperature Protection	Level 1: 80C Delay: N/A alarm only Recovery: N/A Level 2: 85C Delay: 500 mS Recovery: 80 C Level 3: 90C Delay: 500 ms Recovery: 80 C
Cell Inbalance Protection	Level 1: 0.3V Delay: N/A alarm only Recovery: N/A Level 2: 0.5V Delay: 500 mS Recovery: 0.2V Level 3: 0.7V Delay: 500 ms Recovery: 0.2V .

BATTERY PACK CONNECTION + OPERATION

The 15EB battery pack utilizes a Amphenol Duraswap Battery SOC Cable Assy. This is a unidirectional blind mate connector. The connector includes 2 power poles and 6 signal poles. The power poles are marked with (+) for positive and (-) for the negative. The 6 data poles are marked with 1 + 2 and 5 + 6. Poles 3 + 4 are not marked but can be found by referencing pole 3 closest to pole 1 + 2 and pole 4 closest to 5 + 6. Poles are used as following



- 1) CAN High
- 2) CAN Low
- 3) Aux thermistor
- 4) Aux thermistor
- 5) Key Switch +
- 6) Key Switch -

The battery utilizes a key switch which must be closed for the battery to charge or discharge. BMS remains active even when the key switch is open.

The battery will go into sleep mode (<800 uA consumption) function if the battery is inactive for >60 minutes when there is no communication, no charge/discharge current, and no wake-up signal. During sleep mode, the charging and discharging MOSFETs remain connected unless the battery undervoltage is detected, at which point the MOSFETs will disconnect. If the BMS detects communication signals or charge/discharge currents ($\geq 2A$, and for charging activation, the charger's input voltage must be at least 2V higher than the battery voltage, or there is a wake-up signal), it will immediately respond and enter the wake-up working state.

The battery has an auxiliary thermistor for easy temperature monitoring. It uses a 10K NTC thermistor B value = 3950.

Batteries are specified with 2 CAN address (1 + 2) The address can be determined by the serial number affixed to the battery case or through the broadcasted CAN information (see below). The last 2 digits in the serial number indicate the CAN address ID

BMS CAN BUS COMMUNICATION PROTOCOL

The battery communicates over CAN bus using extended (29-bit) frame IDs. Send a request frame with an 8-byte all-zero payload and the BMS respond with the address bytes swapped and the data payload.

CAN Bus Settings:

Parameter	Value
Baud Rate	250 kbps
Frame Type	Extended (29-bit ID)
Payload	8 bytes per frame

Default Addresses:

Role	Address
Host	0x40
Battery Unit 1	0x01
Battery Unit 2	0x02

Frame ID Format:

Request (Host → Battery) : **0x18 <DataID> <BatteryAddr> <HostAddr>**

Example : **0x18900240**
 0x18 = priority
 0x90 = data ID
 0x02 = battery address (0x01 or 0x02)
 0x40 = host address

Response (Battery → Host) : **0x18 <DataID> <HostAddr> <BatteryAddr>**

Example : **0x18904002**
 0x18 = priority
 0x90 = data ID
 0x40 = host address
 0x02 = battery address (0x01 or 0x02)

BMS CAN DATA ID REFERENCE

0x90 - Pack Totals (Voltage, Current, SoC)

Byte	Field	Encoding
0 - 1	Pack voltage	uint16 Big Endian, units of 0.1V
2 - 3	Reserved	-
4 - 5	Pack current	uint16 Big Endian, offset 30000, units of 0.1A
6 - 7	State of charge	uint16 Big Endian, units of 0.1%

Current decoding: $\text{current_A} = (\text{raw_value} - 30000) \times 0.1$. Values >30000 = charging; <30000 = discharging; 30000 = 0.0A.

0x91 - Cell Voltage Extremes

Byte	Field	Encoding
0 - 1	Maximum cell voltage	uint16 Big Endian, units of 1mV
2	Cell # with max voltage	uint8 (1-based)
3 - 4	Minimum cell voltage	uint16 Big Endian, units of 1mV
5	Cell # with min voltage	uint8 (1-based)
6 - 7	Reserved	-

0x92 - Temperature Extremes

Byte	Field	Encoding
0	Maximum temperature	uint8, subtract 40 for °C
1	Sensor # with max temp	uint8 (1-based)
2	Minimum temperature	uint8, subtract 40 for °C
3	Sensor # with min temp	uint8 (1-based)
4 - 7	Reserved	-

0x93 - MOS State and Capacity

Byte	Field	Encoding
0	State	0 = idle, 1 = charging, 2 = discharging
1	Charge MOS	0 = off (blocked), 1 = on (allowed)
2	Discharge MOS	0 = off (blocked), 1 = on (allowed)
3	BMS life counter	uint8 (0-255, wrapping)
4 - 7	Remaining capacity	uint32 Big Endian, units of mAh

0x94 - Status Information

Byte	Field	Encoding
0	Number of cells	uint8

1	Number of temp sensors	uint8
2 - 7	Reserved	-

0x95 - Individual Cell Voltages (Multi-Frame Response)

Byte	Field	Encoding
0	Frame number	uint8, 1-based (frame 1 = cells 1-3, frame 2 = cells 4-6, etc.)
1 - 2	Cell 1 voltage	uint16 Big Endian, units of 1mV
3 - 4	Cell 2 voltage	uint16 Big Endian, units of 1mV
5 - 6	Cell 3 voltage	uint16 Big Endian, units of 1mV
7	Reserved	-

For a 13-cell pack: send one request and read up to 13 cells only. Discard values beyond that index.

0x96 - Individual Temperatures

Byte	Field	Encoding
0	Frame number	uint8
1 - 7	Temperature values	uint8 each, subtract 40 for °C

For a 4-sensor pack: send 1 request and read bytes 1-4 only.

0x97 - Cell Balance State

Byte	Field	Encoding
0	Cells 1-8 balance mask	uint8 bitmask (bit 0 = cell 1)
1	Cells 9-16 balance mask	uint8 bitmask
2	Cells 17-24 balance mask	uint8 bitmask
3 - 7	Reserved	-

Bit set = cell is actively balancing.

0x98 - Fault Status (8 bytes)

Some faults have two levels: Alarm (warning only) and Protection (MOS open, charge/discharge disabled).

Byte	Bit	Fault
0	0	Cell overvoltage alarm
	1	Cell overvoltage protection
	2	Cell undervoltage alarm
	3	Cell undervoltage protection
	4	Pack overvoltage alarm
	5	Pack overvoltage protection
	6	Pack undervoltage alarm

	7	Pack undervoltage protection
1	0	Charge over-temperature alarm
	1	Charge over-temperature protection
	2	Charge under-temperature alarm
	3	Charge under-temperature protection
	4	Discharge over-temperature alarm
	5	Discharge over-temperature protection
	6	Discharge under-temperature alarm
	7	Discharge under-temperature protection
2	0	Charge overcurrent alarm
	1	Charge overcurrent protection
	2	Discharge overcurrent alarm
	3	Discharge overcurrent protection
	6	SOC low alarm level 1
	7	SOC low alarm level 2
3	0	Cell voltage difference alarm level 1
	1	Cell voltage difference alarm level 2
	2	Temperature difference alarm level 1
	3	Temperature difference alarm level 2
	4-7	Reserved
4	0	Charge MOS over-temperature protection
	1	Discharge MOS over-temperature protection
	2-7	BMS Internal fault
5 - 7	all	BMS Internal fault

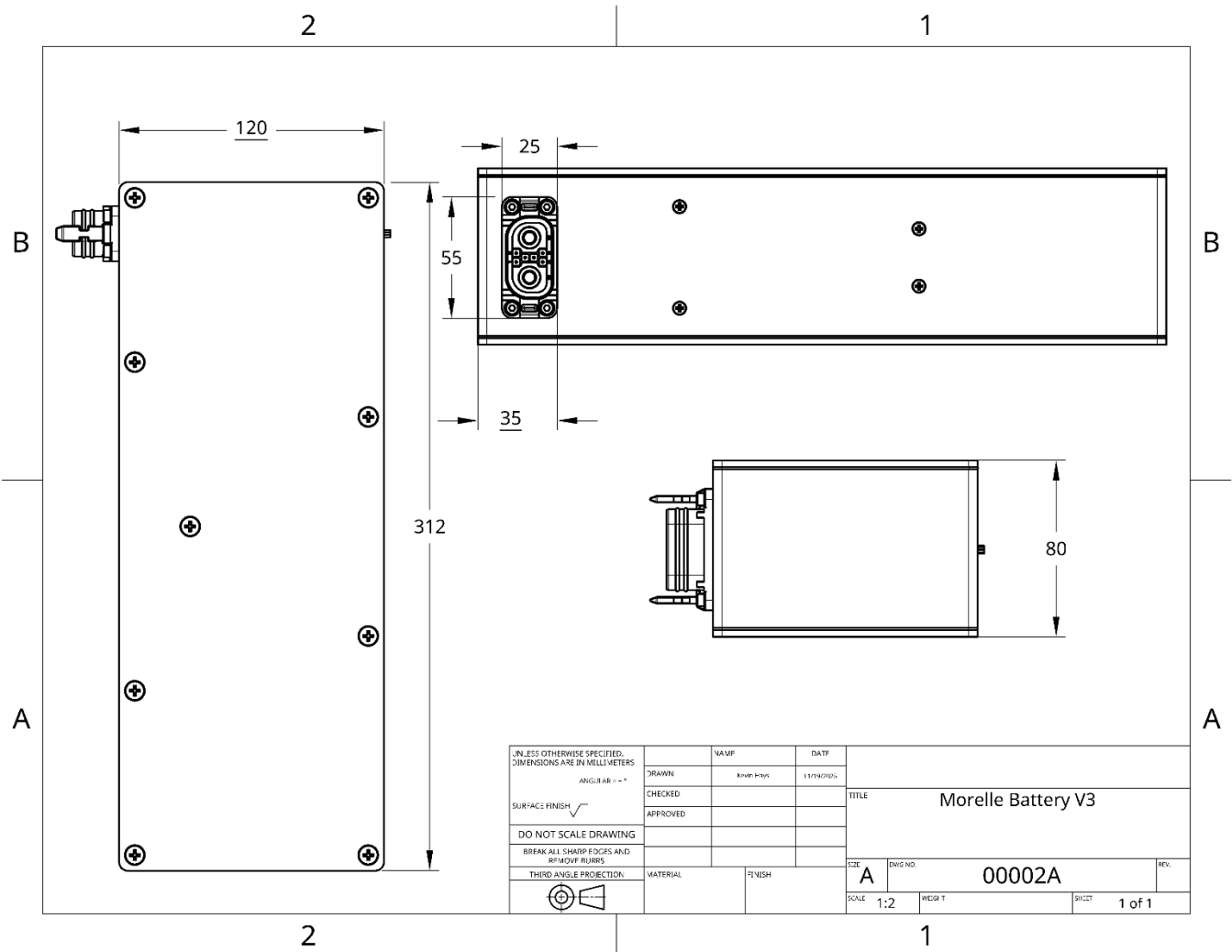
SAFETY INFORMATION

- This battery pack is designed for use with the Morelle Charger. Do not charge with other chargers. The BMS protection parameters are calibrated for the specific cell configuration in your pack.
- The BMS automatically disables the charge or discharge MOS if unsafe conditions are detected (protection).
- The BMS automatically disables if the key switch pins are not closed
- Keep the battery within its operating temperature range (-10°C ~ 60°C).
- Do not open the battery enclosure. No user-serviceable parts inside.
- The Top and bottom aluminum plates utilize a dielectric powder coat. Avoid scratching or damaging this coating to maintain insulation.
- The Top and bottom plates may get warm during charge/discharge. This is normal and care should be taken if handling the battery
- When battery packs are used in parallel, packs which have similar voltage should be utilized to avoid inrush current from different charge level.
- The battery includes an internal High current fuse, if this fuse blows the battery must be returned for evaluation + servicing

TROUBLESHOOTING QUICK REFERENCE

Problems	Steps
Battery not detected by charger	1. Verify the battery is connected to the charger 2. Check the CAN bus cable connection 3. Power cycle the charger
Battery reporting faults (0x98 non-zero)	1. Check the 0x98 fault status bytes to identify the specific fault 2. Alarm faults are warnings only, the battery continues operating 3. Protection trigger MOS disabled, the condition must clear before the battery re-enables the affected MOS
CAN communication not working	1. Verify CAN bus wiring and termination 2. Confirm 250 kbps baud rate and extended (29-bit) frame IDs 3. Check that the correct battery address is being used in request frames

DRAWING



SUPPORT

For technical support or inquiries:

hello@morelle.ai

Version History

Version	Date	Author	Notes
1.0	5/19/2026	KH/ID	